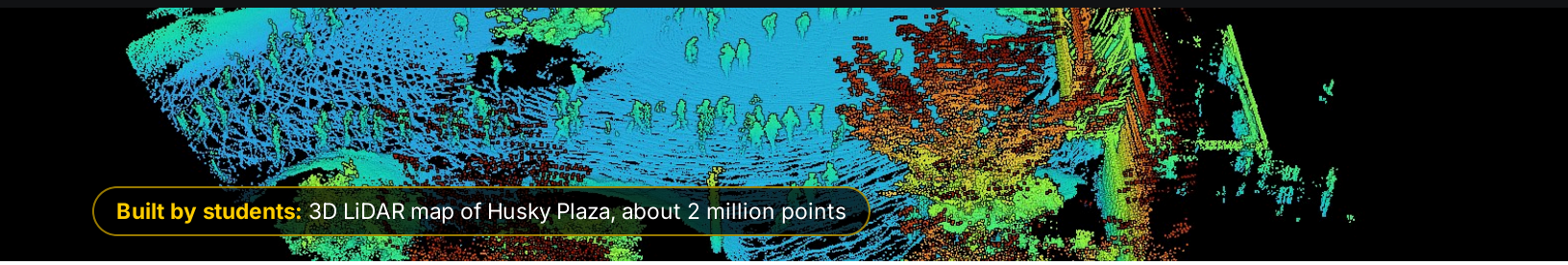
MICHIGAN TECH · SPRING 2027

Build the robot brain.



EE5531 Introduction to Robotics

Take a robot from **raw sensor data** to **driving itself**. You'll localize, map, and navigate real robots with **ROS 2** and **Nav2**, the same stack used on autonomous vehicles and robots in industry.



Built by students: 3D LiDAR map of Husky Plaza, about 2 million points

Real robots, every week

Clearpath Jackals and TurtleBot3s with LiDAR and IMUs. Not just simulation.

The industry toolchain

ROS 2, Nav2, SLAM, Gazebo, RViz2, Python, and Git: the skills in robotics and AV job postings.

The math behind autonomy

Kalman filters, sensor models, mapping, path planning, and behavior trees.

A portfolio you can show

Public GitHub repos with maps, videos, and interactive 3D results.

Jackal Patrol Challenge Your capstone: program a Jackal to patrol autonomously, then **compete**. Performance is scored.

— WHERE EE5531 ALUMNI WORK NOW —

MAY mobility · TORC · gm general motors · TORO · Qualcomm

...and throughout the autonomous vehicle sector.

LECTURE	Mon & Wed, 1:00–1:50 PM
PROJECTS	Mon, 2:00–3:50 PM
CRN	12485 / 12486
PREREQS	FOSS101 or equivalent experience
FOR	MTU Robotics Engineering BS grads, and anyone with Linux, Python/C++, and Git experience
CONTACT	Dr. Jeremy Bos · jpbos@mtu.edu

Runs only with enough enrollment. Register early and bring a friend.

Scan for the full course site

robust-autonomous-systems-laboratory.github.io/ee5531